

Practical: - 8

Aim: - To interface RFID module with Raspberry Pi and to read and write to RFID tags/ cards

RFID stands for **Radio Frequency Identification**. It is a wireless technology that uses radio waves to automatically identify, track, and communicate data between a tag and a reader.

How RFID Works

- **Tags:** Small microchips with antennas attached to objects to store data.
 - **Readers:** Devices that send out radio waves to activate the tags and read their stored information.
 - **No Line of Sight:** Unlike standard barcodes, RFID does not need a direct line of sight to be scanned
- Uses:** Common in retail inventory, logistics, access cards, and animal tracking.

Radio Waves:

Radio waves are a type of electromagnetic radiation that have the longest wavelengths and the lowest frequencies in the electromagnetic spectrum.

Key Characteristics

- **Wavelength:** They stretch from 1 millimeter to over 100 kilometers long.
- **Frequency:** They range from 3 kilohertz (kHz) to 300 gigahertz (GHz). A frequency is how many times a wave repeats in one second.
- **Speed:** They travel at the speed of light in a vacuum

Common Uses

- **Broadcasting:** AM and FM radio stations send music and news.
- **Communication:** Cell phones and Wi-Fi networks use them to share information.
- **Navigation:** Air traffic control and GPS systems use them to track locations

Part I: - Connection

1. RFID Connection with Raspberry pi

GND = Pin no 6

VCC (3V) = Pin no 1 (**This Device works on 3V**)

SDA (Serial Data) = Pin no 3

SCL (Serial Clock Line) = Pin no 5

2. LED Connection

LED = Pin no 11 (GPIO 17)

GND = Pin no 9

Part II – Library Installation

Open the Terminal: -

- sudo apt update

```
surya@raspberrypi:~$ sudo apt update
Hit:1 http://raspbian.raspberrypi.com/raspbian bookworm InRelease
Hit:2 http://archive.raspberrypi.com/debian bookworm InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
420 packages can be upgraded. Run 'apt list --upgradable' to see them.
# http://raspbian.raspberrypi.com/raspbian/dists/bookworm/InRelease: Key
) for details.
surya@raspberrypi:~$
```

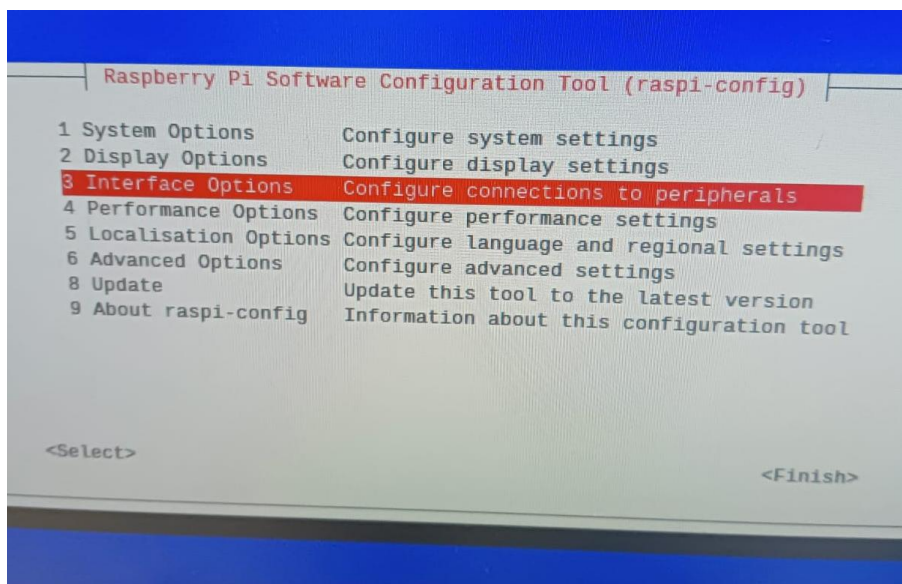
- sudo apt install -y python3-pip python3-venv python3-libgpiod-dev python3-libgpiod

```
surya@raspberrypi:~$ sudo apt install -y python3-pip python3-venv python3-libgpiod-dev python3-libgpiod
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package python3-venv
surya@raspberrypi:~$
```

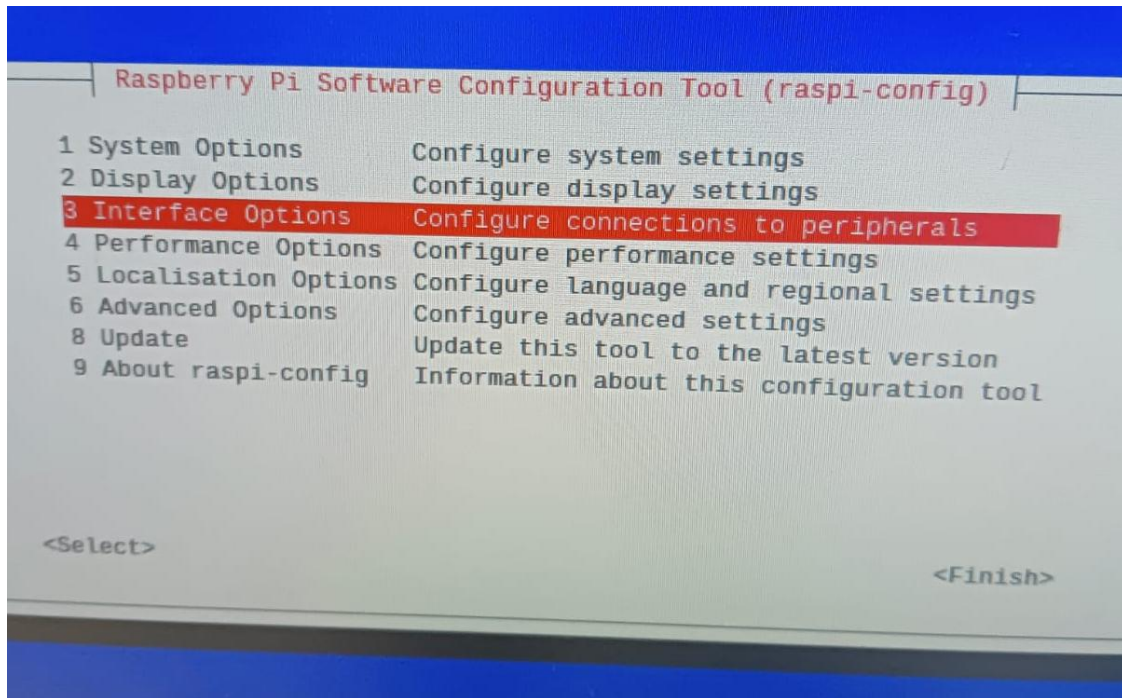
- sudo raspi-config

```
surya@raspberrypi:~$ sudo raspi-config
surya@raspberrypi:~$
```

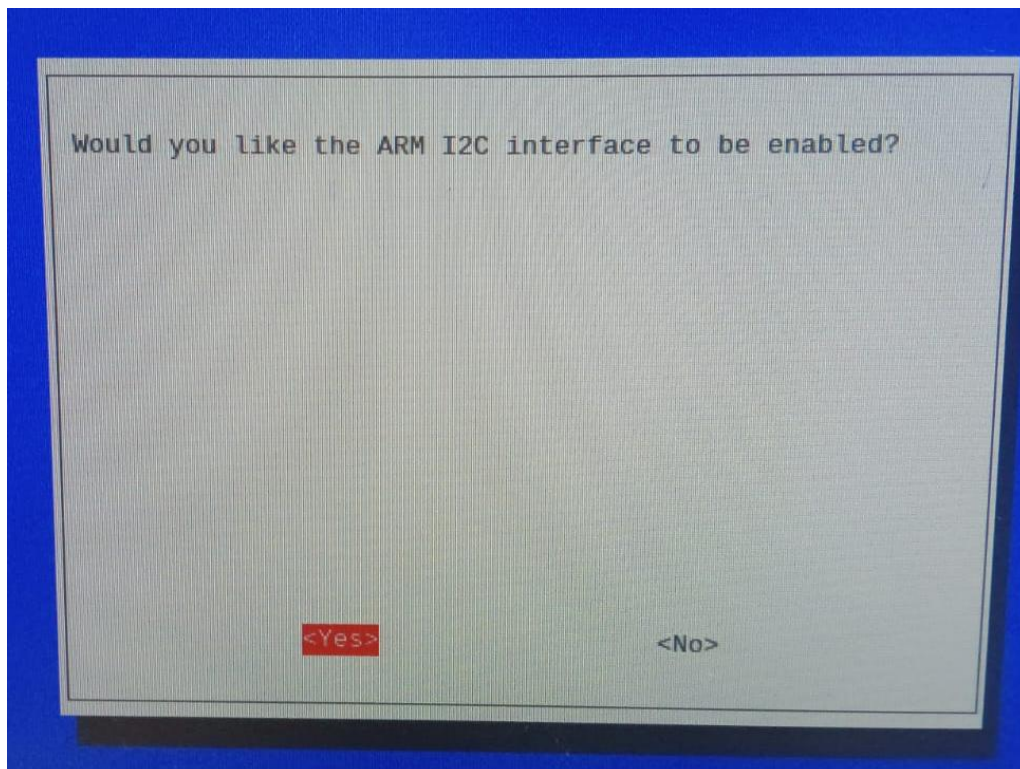
- Click on → “Interface Options”



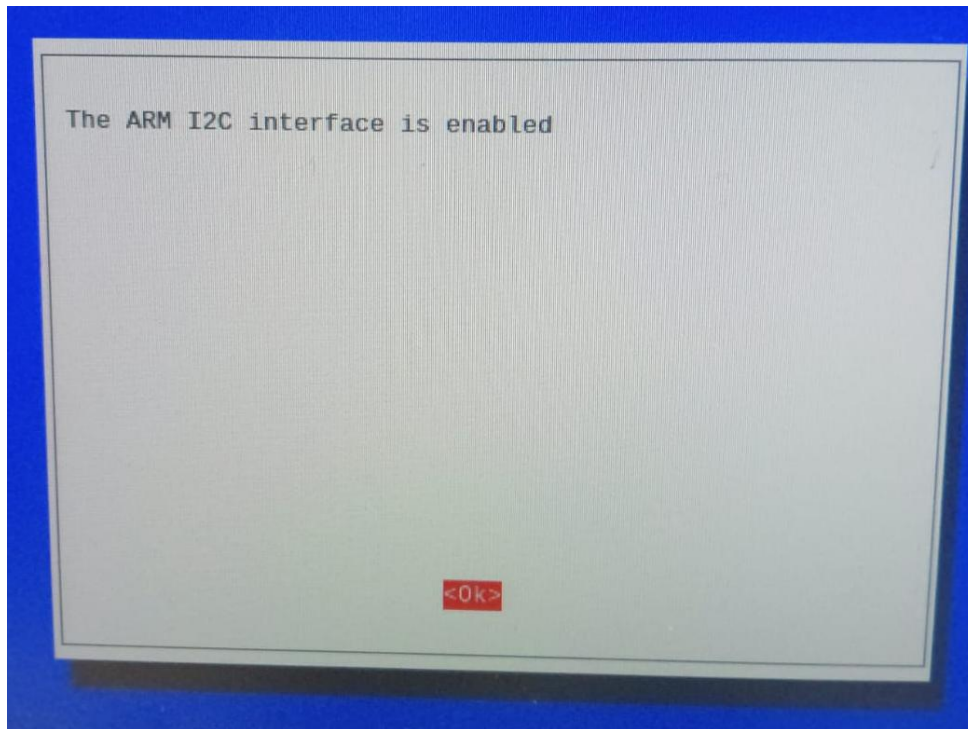
- Click on → “i2c”



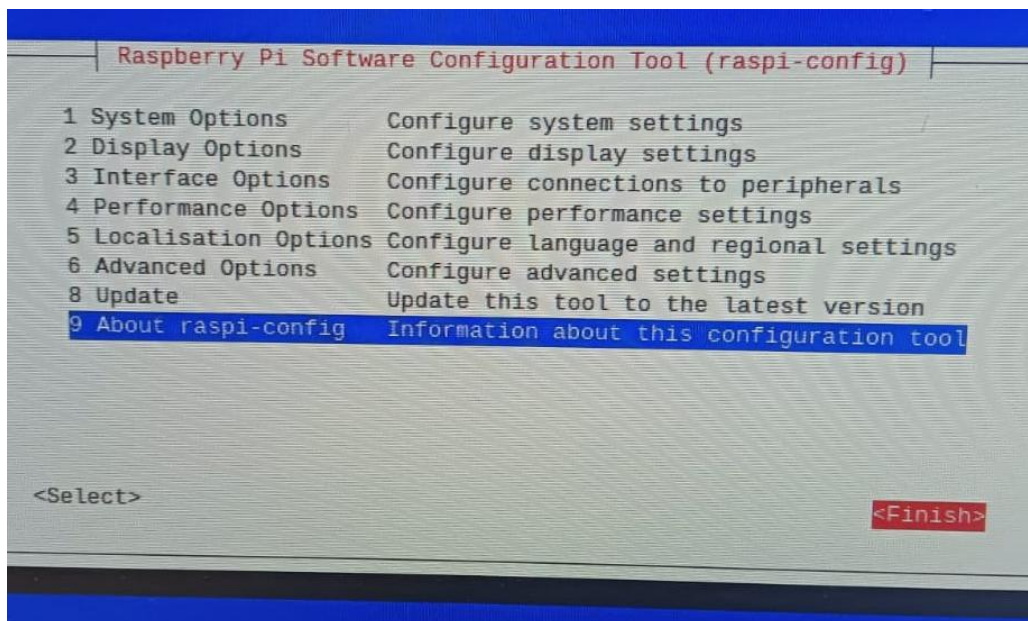
- Click on -> “Yes”



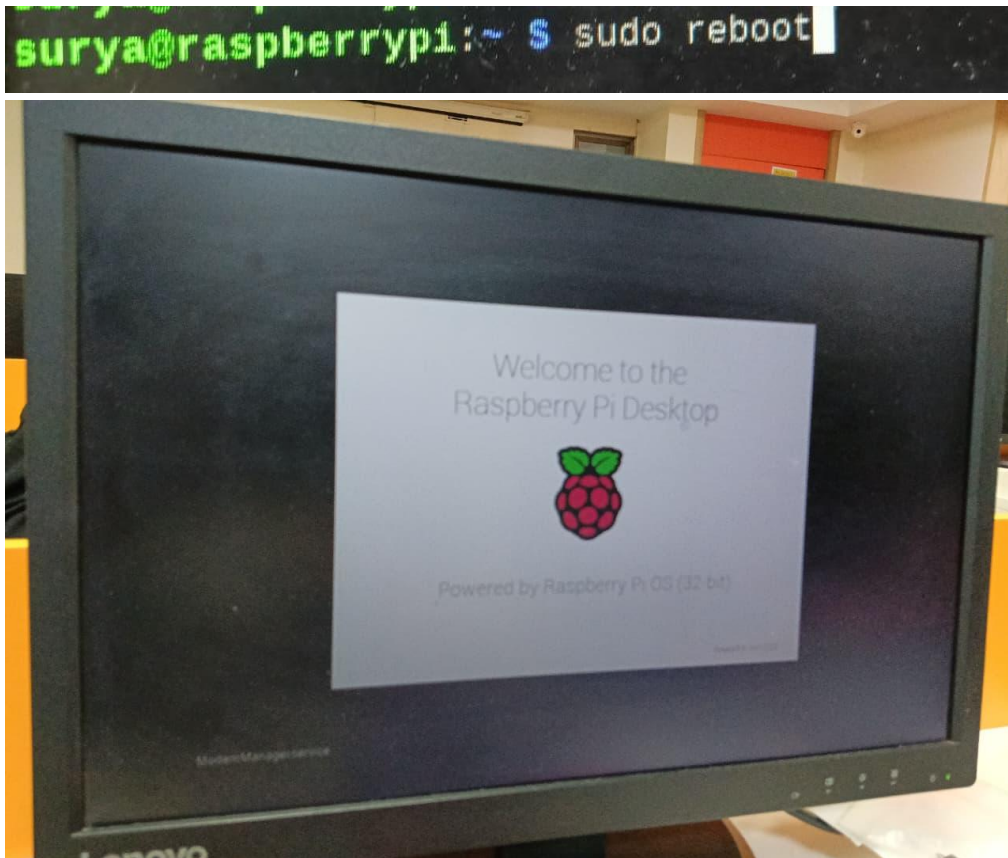
- Click on → “Ok”



- Click on → “Finish”



- `sudo reboot`



- `ls /dev/i2c*`

```
surya@raspberrypi:~ $ ls /dev/i2c*  
/dev/i2c-1 /dev/i2c-13 /dev/i2c-14  
surya@raspberrypi:~ $
```

- i2cdetect -y 1

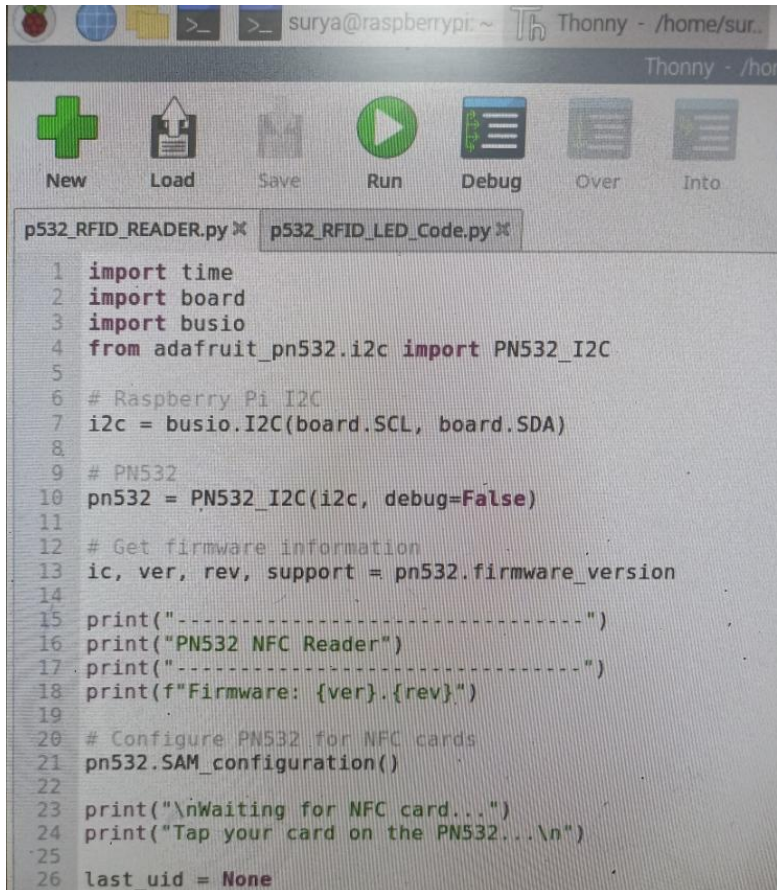
```
surya@raspberrypi1:~$ i2cdetect -y 1
    0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
30:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
10:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
20:  --  --  --  --  24  --  --  --  --  --  --  --  --  --  --  --
30:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
40:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
50:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
60:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
70:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
surya@raspberrypi1:~$
```

- pip3 install adafruit-blinka adafruit-circuitpython-pn532 --break-system-packages

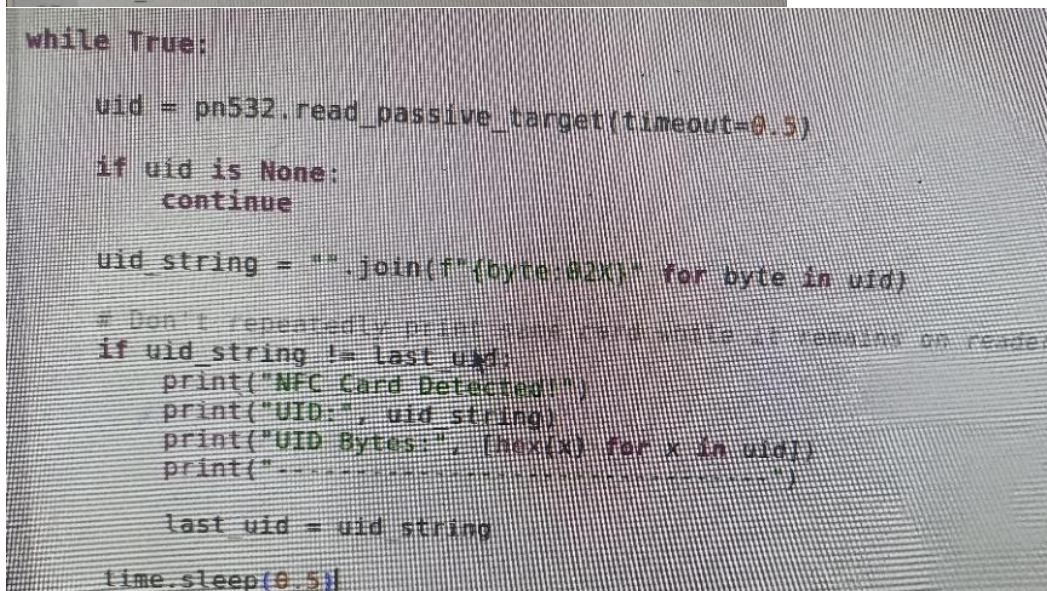
```
surya@raspberrypi1:~$ pip3 install adafruit-blinka adafruit-circuitpython-pn532 --break-system-packages
Discarding https://www.piwheels.org/simple/adafruit-circuitpython-connectionmanager/adafruit_circuitpython_connectionmanager-3.1.8-py3-none-any.whl.metadata (4.5 kB)
https://www.piwheels.org/simple/adafruit-circuitpython-connectionmanager/adafruit_circuitpython_connectionmanager-3.1.8-py3-none-any.whl.metadata: Requested Adafruit-Circuitpython-ConnectionManager from adafruit-circuitpython-requests->adafruit-circuitpython-typing->adafruit-blinka has inconsistent Name: expected 'Adafruit-Circuitpython-ConnectionManager', but metadata has 'adafruit-circuitpython-connectionmanager'
Downloading adafruit_circuitpython_connectionmanager-3.1.8-py3-none-any.whl (7.8 kB)
Downloading https://www.piwheels.org/simple/adafruit-circuitpython-pn532/adafruit_circuitpython_pn532-2.4.9-py3-none-any.whl (15 kB)
Downloading https://www.piwheels.org/simple/adafruit-platformdetect/adafruit_platformdetect-3.89.1-py3-none-any.whl (27 kB)
Downloading https://www.piwheels.org/simple/adafruit-pureio/Adafruit_PureIO-1.1.11-py3-none-any.whl (10 kB)
Downloading https://www.piwheels.org/simple/binho-host-adapter/binho_host_adapter-0.1.6-py3-none-any.whl (10 kB)
Downloading https://www.piwheels.org/simple/sysv_ipc/sysv_ipc-1.2.0-cp311-cp311-linux_armv7l.whl (66 kB)
67.0/67.0 kB 40.0 kB/s eta 0:00:00
Downloading https://www.piwheels.org/simple/adafruit-circuitpython-busdevice/adafruit_circuitpython_busdevice-5.2.17-py3-none-any.whl (7.5 kB)
Downloading https://www.piwheels.org/simple/adafruit-circuitpython-typing/adafruit_circuitpython_typing-1.12.3-py3-none-any.whl (11 kB)
Downloading https://www.piwheels.org/simple/pyusb/pyusb-1.3.1-py3-none-any.whl (58 kB)
58.5/58.5 kB 191.3 kB/s eta 0:00:00
Using cached https://www.piwheels.org/simple/adafruit-circuitpython-requests/adafruit_circuitpython_requests-4.1.17-py3-none-any.whl (10 kB)
Using cached https://www.piwheels.org/simple/adafruit-circuitpython-pn532/adafruit_circuitpython_pn532-2.4.9-py3-none-any.whl (15 kB)
Using cached https://www.piwheels.org/simple/adafruit-platformdetect/adafruit_platformdetect-3.89.1-py3-none-any.whl (27 kB)
Using cached https://www.piwheels.org/simple/adafruit-pureio/Adafruit_PureIO-1.1.11-py3-none-any.whl (10 kB)
Using cached https://www.piwheels.org/simple/binho-host-adapter/binho_host_adapter-0.1.6-py3-none-any.whl (10 kB)
Using cached https://www.piwheels.org/simple/sysv_ipc/sysv_ipc-1.2.0-cp311-cp311-linux_armv7l.whl (66 kB)
Using cached https://www.piwheels.org/simple/adafruit-circuitpython-busdevice/adafruit_circuitpython_busdevice-5.2.17-py3-none-any.whl (7.5 kB)
Using cached https://www.piwheels.org/simple/adafruit-circuitpython-typing/adafruit_circuitpython_typing-1.12.3-py3-none-any.whl (11 kB)
Using cached https://www.piwheels.org/simple/pyusb/pyusb-1.3.1-py3-none-any.whl (58 kB)
Using cached https://www.piwheels.org/simple/adafruit-circuitpython-requests/adafruit_circuitpython_requests-4.1.17-py3-none-any.whl (10 kB)
Using cached https://www.piwheels.org/simple/adafruit-circuitpython-pn532/adafruit_circuitpython_pn532-2.4.9-py3-none-any.whl (15 kB)
Installing collected packages: sysv_ipc, Adafruit-PlatformDetect, pyusb, binho-host-adapter, Adafruit-PureIO, pyftdi, Adafruit-Circuitpython-ConnectionManager, adafruit-blinka, adafruit-circuitpython-requests, adafruit-circuitpython-typing, adafruit-circuitpython-busdevice, adafruit-circuitpython-pn532
Successfully installed Adafruit-Circuitpython-ConnectionManager-3.1.8 Adafruit-PlatformDetect-3.89.1 Adafruit-PureIO-1.1.11 adafruit-blinka-9.2.0 adafruit-circuitpython-busdevice-5.2.17 adafruit-circuitpython-pn532-2.4.9 adafruit-circuitpython-requests-4.1.17 adafruit-circuitpython-typing-1.12.3 binho-host-adapter-0.1.6 pyftdi-0.97.2 pyusb-1.3.1 sysv_ipc-1.2.0
surya@raspberrypi1:~$
```

Part III – Coding /program

1. RFID READER



```
p532_RFID_READER.py x p532_RFID_LED_Code.py x
1 import time
2 import board
3 import busio
4 from adafruit_pn532.i2c import PN532_I2C
5
6 # Raspberry Pi I2C
7 i2c = busio.I2C(board.SCL, board.SDA)
8
9 # PN532
10 pn532 = PN532_I2C(i2c, debug=False)
11
12 # Get firmware information
13 ic, ver, rev, support = pn532.firmware_version
14
15 print("-----")
16 print("PN532 NFC Reader")
17 print("-----")
18 print(f"Firmware: {ver}.{rev}")
19
20 # Configure PN532 for NFC cards
21 pn532.SAM_configuration()
22
23 print("\nWaiting for NFC card...")
24 print("Tap your card on the PN532...\n")
25
26 last_uid = None
```



```
while True:
    uid = pn532.read_passive_target(timeout=0.5)
    if uid is None:
        continue
    uid_string = "".join(f"{byte:02X}" for byte in uid)
    # Don't repeatedly print same card while it remains on reader
    if uid_string != last_uid:
        print("NFC Card Detected!")
        print("UID:", uid_string)
        print("UID Bytes:", (hex(x) for x in uid))
        print("-----")
        last_uid = uid_string
    time.sleep(0.5)
```

2. RFID LED CODE

```
p532_RFID_READER.py x p532_RFID_LED_Code.py x
1 import time
2 import board
3 import busio
4 import digitalio
5 from adafruit_pn532.i2c import PN532_I2C
6
7 # =====
8 # SETTINGS
9 # =====
10
11 AUTHORIZED_UID = "B6339904"
12
13 LED_PIN = board.D17
14
15 # =====
16 # LED SETUP
17 # =====
18
19 led = digitalio.DigitalInOut(LED_PIN)
20 led.direction = digitalio.Direction.OUTPUT
21 led.value = False
22
23 # =====
24 # PN532 SETUP
25 # =====
26
27 i2c = busio.I2C(board.SCL, board.SDA)
28
29 pn532 = PN532_I2C(i2c, debug=False)
30
31 ic, ver, rev, support = pn532.firmware_version
32
33 print("-----")
34 print("PN532 Access Control")
35 print("-----")
36 print(f"PN532 Firmware: {ver}.{rev}")
37
38 pn532.SAM_configuration()
39
40 print("Waiting for NFC card...\n")
41
42 led_state = False
43 last_uid = None
44
45 # =====
46 # MAIN LOOP
47 # =====
48
49 while True:
50
51     uid = pn532.read_passive_target(timeout=0.5)
52
53     if uid is None:
54         continue
55
56     uid_string = "".join(f"{byte:02X}" for byte in uid)
57
58     # Avoid reading same card continuously
```

p532_RFID_READER.py ✕

p532_RFID_LED_Code.py ✕

```
56     uid_string = "".join("{}{byte:02X}" for byte in uid)
57
58     # Avoid reading same card continuously
59     if uid_string == last_uid:
60         continue
61
62     last_uid = uid_string
63
64     print("Card detected!")
65     print("UID:", uid_string)
66
67     # =====
68     # UID AUTHENTICATION
69     # =====
70
71     if uid_string == AUTHORIZED_UID:
72
73         print("ACCESS GRANTED")
74
75         # Toggle LED
76         led_state = not led_state
77         led.value = led_state
78
79         if led_state:
80             print("LED: ON")
81         else:
82             print("LED: OFF")
83
84     else:
85
86         print("ACCESS DENIED")
87         print("Unknown card!")
88
89     print("-----")
90
91     time.sleep(1)
92
93     # Allow same card to be detected again
94     last_uid = None
95
```

Part IV: - Output

